Supplementary Methods

Drug tolerance takes shape: persistent homology reveals cyclic cell-state plasticity in EGFR-mutant lung cancer

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S1. Patient-atlas filtering details

For GSE131907 [1] we filtered to cells with `Sample_Origin == 'tLung'` (primary lung tumor) AND `Cell_type` containing 'Epithelial', using the cell-type annotations from `GSE131907_Lung_Cancer_cell_ann` From the resulting 7,270 tumor epithelial cells, a uniform random sample of 5,000 cells was drawn (`random_state=42`), then 1,500 cells were subsampled further for tractable Vietoris–Rips computation.

S2. Highly-variable gene (HVG) batch keys

HVG selection used `batch_key='timepoint'` for GSE134839 and GSE150949 (samples share a clonal PC9 background but differ in drug-exposure timepoint, treated as a technical batch following Luecken and Theis [2]), `batch_key='pdx'` for GSE243562 (to account for inter-PDX-model variability), and `batch_key='Sample'` for the Kim atlas (per-patient variability).

S3. PDX mouse-gene filter

For PDX data, we filtered mouse genes using the pattern `r'^mm10.*|^[a-z][a-z0-9]+$'` applied to gene symbols (matching either the Ensembl mouse prefix `mm10` or lowercase gene symbols characteristic of MGI-style mouse annotations). Of 82,350 initial features, 24,062 mouse-like genes (29%) were removed, leaving 58,288 human-like features. We acknowledge this symbol-based heuristic is inferior to alignment-based xeno-classification (e.g. XenoCell, Xenomake) for publication-grade PDX analysis; our downstream results depend on PCA structure and are insensitive to modest filter errors, but this is flagged as a limitation in the main text Discussion.

S4. Max H_1 as a hybrid statistic

Max H_1 is a hybrid statistic: the existence of a nonzero H_1 bar is a topological invariant (a nontrivial 1-cycle exists), but the numerical value of its lifespan is measured in filtration-parameter units (here, Euclidean distance in PCA space) and is therefore metric-dependent. We refer to the statistic as "topological" because it reports on the presence and robustness of a topological feature, while acknowledging that the numerical scale depends on the metric used. The PC-count sensitivity sweep (S5) quantifies the metric dependence.

S5. PC-count sensitivity

Max H_1 was computed on the D9 erlotinib timepoint ($n = 379$) and D14 osimertinib timepoint ($n = 1{,}000$) at $n_{\mathrm{PCs}} \in \{10, 20, 30, 50\}$. Values were 1.96/2.08/1.77/1.03 for D9 and

3.12/5.56/3.78/4.40 for D14. The ordering drug-treated > untreated is preserved at every PC count, but absolute values vary within a factor of 2. The $n_{\mathrm{PCs}} = 30$ choice used throughout was selected for consistency with prior scTDA work [3]. See also Supplementary Table S5.

S6. EMT-cassette gene list (Mapper filter)

EMT score (used as Mapper filter for the cell-line analysis) was computed via `scanpy.tl.score_genes` on a 33-gene MSigDB Hallmark EMT cassette: *VIM*, *CDH2*, *FN1*, *SNAI1*, *SNAI2*, *ZEB1*, *ZEB2*, *TWIST1*, *MMP2*, *MMP9*, *CDH1*, *OCLN*, *TJP1*, *DSP*, *KRT19*, *KRT8*, *SERPINE1*, *TGFBI*, *SPARC*, *COL1A1*, *COL3A1*, *COL5A2*, *ACTA2*, *TAGLN*, *MYL9*, *CALD1*, *LGALS1*, *TPM1*, *TPM2*, *FSTL1*, *ITGB1*, *LAMC2*, *LOXL2*. Of these, 28 of 33 genes were present in the GSE134839 normalized expression matrix and used for scoring.

S7. Harmony batch-correction parameters

Harmony [4] was applied to the top-50 PCA components with default parameters: `theta = 2`, `lambda = 1`, `nclust = 100`, `epsilon_cluster = 1e-5`, `epsilon_harmony = 1e-4`, `max_iter_harmony = 10`, `max_iter_kmeans = 20`. Convergence was reached in 2 iterations for the osimertinib cohort. Batch keys: `timepoint` for GSE150949, per-sample GSM identifier for GSE243562, per-patient `Sample` identifier for GSE131907.

S8. Pipeline architecture

The framework is organised as numbered modules, each accepting a standard `AnnData` object and producing typed intermediate outputs (persistence diagrams as `.npy`, Mapper graphs as pickled dicts, enrichment results as `.csv`):

- `01_preprocess.py`: load GEO data, QC, normalise, HVG, regress, PCA, cell-cycle score.
- `02_standard_analysis.py`: UMAP, Leiden clustering [5] (multi-resolution), marker genes, EMT score, diffusion map, PAGA.
- `03_topology.py`: per-group persistent homology, Mapper (3 filters), pairwise Wasserstein, permutation tests.
- `04_cell_cycle_ablation.py`: remove cell-cycle genes, re-do PCA and PH, compute preservation ratio + permutation null.
- `05_gene_attribution.py`: Mapper gene connectivity, differential expression, pathway enrichment.
- `06_validation.py`: apply the full pipeline to a validation dataset, compute cross-dataset Wasserstein.
- `07_figures.py`: assemble publication figures from saved intermediate outputs.
- `08_validation_pdx.py`: PDX-specific loader and persistent homology for the ASCL1 PDX cohort.
- `09_validation_kim_atlas.py`: Kim 2020 patient atlas baseline loading and H_1 computation.
- `10_ablation_permtest.py`: extended permutation tests on cell-cycle-ablated data.
- `11_pdx_gene_attribution.py`: cross-system gene-overlap analysis between cell line and PDX.

- `12_convert_tables_to_parquet.py`: CSV → Parquet for large tables.
- `13_harmony_batch_correction.py`: Harmony batch-correction sensitivity analysis.
- `14_validation_maynard.py`: Maynard 2020 neo-osi cohort loader.
- `15_maynard_full_cohort.py`: full Maynard 14-patient EGFR-mutant cohort analysis.

Users can substitute their own datasets by providing an `AnnData` file with a `timepoint` (or equivalent) observation column.

S9. Unit tests

The `sctda_plasticity` package includes pytest-based unit tests covering persistent homology on synthetic circle and cluster data, Mapper graph construction with known topology, gene connectivity ranking, and permutation test structure. All 8 tests pass on the submitted version; see `tests/test_topology.py`. Code is lint-clean under `ruff check` with a 100-character line limit.

S10. Resampling stability of max H_1

To characterise max H_1 variability under subsampling, we ran the 1,000-cell subsample five times with independent random seeds for four cohorts (osi D14, osi D0, PDX YU-006 residual, PDX YU-006 untreated). Coefficients of variation (CV = SD/mean) were 8.9%, 12.4%, 12.8%, and 23.5% respectively. The *ordering* (D14 > D0; residual > untreated) was preserved in every resample. Values in main-text Table 2 (validation cohort) are from seed 42; mean ± SD across the five seeds is in Supplementary Table S7.

S11. Alternative topological summaries

Two alternative summaries of the persistence diagram on the osimertinib time course replicate the monotonic trend. Total H_1 persistence (sum of finite bar lengths) grew 402 → 568 → 588 across D0/D7/D14, while persistence entropy stayed at 6.7–7.0 — indicating the growth is driven by larger-scale features rather than more numerous small ones. All three summaries across ten cohorts: Supplementary Table S6.

S12. Mapper parameter sweep

A 4×5 Mapper-parameter sweep ($n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$; overlap $\in \{0.2, 0.3, 0.4, 0.5\}$) on the cell-line dataset with the EMT filter produced 12–36 Mapper nodes (Table S3). Gene-connectivity rankings were near-identical across all 20 combinations (mean Spearman $\rho = 0.994$ vs. the reference, range 0.990–0.998; top-10 genes reproduced within rank order 1–15; Supplementary Table S8). The downstream EMT-enrichment conclusion is therefore insensitive to the specific Mapper parameter choice.

S13. Stringent S/G2M-score regression detail

Because Tirosh ablation removes only a subset of the cell-cycle program, we also applied the more demanding `sc.pp.regress_out(S_score, G2M_score)` on 1,000-cell subsamples and recomputed max H_1. Across five cohorts (erlotinib D0/D9/D11, PDX YU-006 untreated/residual), the post/pre ratio was 1.07–4.05 (see main-text Table 3; mean 1.97, every value > 1), preserving — and in several cases amplifying — the loop signal after the dominant cell-cycle variance axis is removed. Erlotinib D11 ($n = 249$, ratio 4.05) is a case where regression unmasks topological structure that PCA had obscured. The osimertinib cohort contains zero Tirosh genes in its HVG set (filtered during selection), so the Tirosh-specific contribution to its H_1 signal is ruled out by absence; non-Tirosh cell-cycle genes (e.g. *MKI67*, *TOP2A*) do co-rank in PC1-Mapper, but the converging discovery-cohort + PDX ablation evidence rules out cell-cycle oscillation as the driver.

S14. Computational environment detail

All analyses ran on macOS 15 (arm64) under Python 3.11 in a conda environment specified in `envs/environment.yml`. Key package versions: scanpy 1.11, anndata 0.12, ripser 0.6, persim 0.3, kmapper 2.0, giotto-tda 0.6, scikit-learn 1.3, numpy 1.26, pandas 2.x. The complete environment is reproducible via `conda env create -f envs/environment.yml`. Random seed 42 is used for every stochastic operation (PCA, UMAP, Leiden clustering, subsampling, permutation tests), yielding deterministic outputs on the same platform up to numerical-library nondeterminism (BLAS thread order, OpenMP scheduling). The framework has been tested on both macOS (arm64) and Linux (x86_64).

References

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